

Simulating Loneliness: Human–Generative Agent Comparative Insights

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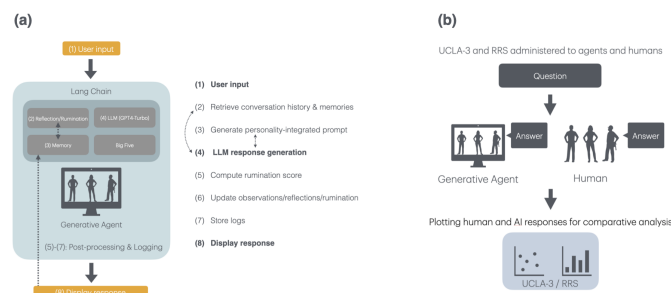


Figure 1: (a) System architecture of the generative agent integrating memory, reflection, Big Five, and LLM response generation, and (b) workflow for administering psychological scales to humans and generative agents.

Abstract

The World Health Organization (WHO) identifies loneliness as a public health risk comparable to smoking or obesity. In this study, we developed generative agents equipped with memory, reflection, and the Big Five personality traits, and compared them with humans using standardized psychological measures: the UCLA Loneliness Scale (UCLA-3) and the Ruminative Responses Scale (RRS). Results revealed a significant positive correlation between loneliness and rumination in both groups ($p \approx 1.7 \times 10^{-9}$), with no significant differences in mean scores or regression slopes. These findings indicate that the agents reproduced the direction and magnitude of the human data, demonstrating the utility of computational models for representing subjective, dynamic internal states in a way directly comparable to standardized measures.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Collaborative and social computing*; • **Computing methodologies** → *Artificial intelligence*.

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Keywords

Generative Agents, Simulation of Loneliness, Human-Computer Interaction, Affective Computing

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1 Introduction

According to the World Health Organization (WHO), loneliness poses a public health risk comparable to smoking or obesity, and is estimated to be associated with 871,000 deaths worldwide annually [World Health Organization 2025].

In this study, we define loneliness as the subjective perception of lacking social connection and note the limitations of conventional agent-based models [Bonabeau 2002; Bourgaïs et al. 2018; Horned and Vanhée 2023] that rely on fixed rules and cannot capture dynamic internal states such as memory and emotion. As an alternative, we propose generative agents [Kambeitz and Meyer-Lindenberg 2025; Park et al. 2023] that integrate memory, reflection, dialogue generation, and Big Five personality traits, and evaluate them against human data [Abe et al. 2025] using the UCLA Loneliness Scale (UCLA-3) and Ruminative Responses Scale (RRS). The results show that the relationship between loneliness and rumination is consistent in both direction and magnitude across agents and

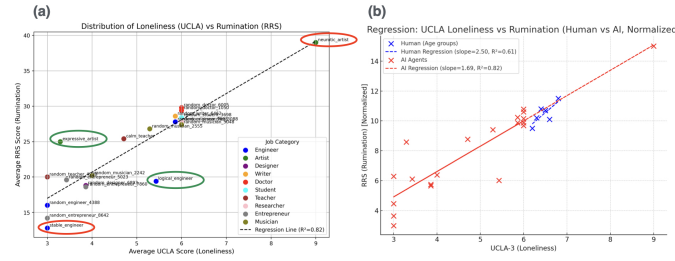


Figure 2: (a) Scatter plot of UCLA-3 vs. RRS for agents with extreme and random Big Five scores. (b) Comparison of normalized regression analyses for humans and agents.

humans, demonstrating the potential of computationally modeling subjective, context-dependent experiences.

2 Method

2.1 Agent Design

The system, implemented in Google Colab using the OpenAI GPT-4-turbo model via the LangChain framework, employs a four-module architecture that ensures behavioral, linguistic, and psychological consistency in simulating loneliness:

- **Memory Module:** Stores short- and long-term memories in a database; retrieves relevant information via Facebook AI Similarity Search for dialogue.
- **Personality Module:** Sets Big Five scores (0–100) to influence behavior and communication style.
- **Reflection Module:** Summarizes recent memories into one- to two-sentence reflections.
- **Dialogue Module:** Produces responses via the OpenAI Application Programming Interface (API), informed by conversation history, memories, and traits.

2.2 Computation and Evaluation of Loneliness and Rumination Scores

The workflow for administering psychological scales to humans and generative agents is shown in Fig. 1(b). Human data were used for comparison, while agents were evaluated with the same UCLA-3 and RRS items automatically. Responses were stored in a database, and mean scores and correlations between loneliness and rumination were compared across groups.

3 Results

Figure 2(a) shows that two agents with extreme Big Five personality scores—namely, *stable_engineer* and *neurotic_artist*—are positioned at opposite ends of the UCLA-3 versus RRS scatter plot (highlighted in red). When additional agents with randomly assigned traits and occupations were overlaid, the coefficient of determination remained high ($R^2 = 0.82$), indicating a strong positive correlation between UCLA-3 scores (x-axis) and RRS scores (y-axis). Even under randomized assignments, the distributions remained closely aligned with the regression line, suggesting that the internal model consistently reproduced the “loneliness–rumination” relationship.

Figure 2(b) compares the normalized regression analyses for humans (blue) and agents (red). Human data exhibited a moderate positive correlation (slope = 2.50, $R^2 = 0.61$), whereas agents showed a stronger correlation (slope = 1.69, $R^2 = 0.82$). In both cases, data points were stably aligned with the regression line. Statistical testing confirmed a significant increase in rumination scores with higher loneliness scores ($p \approx 1.7 \times 10^{-9}$), while neither the mean score difference ($p \approx 0.60$) nor the slope interaction effect ($p \approx 0.75$) was significant. These results indicate that the agents replicated patterns consistent with human data.

4 Conclusion

In this study, we implemented a loneliness simulation using generative agents and demonstrated that the relationship between loneliness and rumination in human data was similarly reproduced. However, the agents do not experience loneliness but answer questionnaire items linguistically based on preset personalities. These findings present a computational framework for modeling loneliness, representing a step toward generating new hypotheses and future studies embedding agents in interactive social contexts.

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